

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA07 **COURSE CODE:** Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

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Code of Conduct:

I B.PARDHU ARJUN(192525245)__(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → _____ → Transport → Network → _____ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3. Given the concept map:

DataLink → MACAddress → LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→ DataLinkLayer → ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✗ → Transport ✗ → Application ✗

Analyze the problem and explain how the concept map helps in troubleshooting.

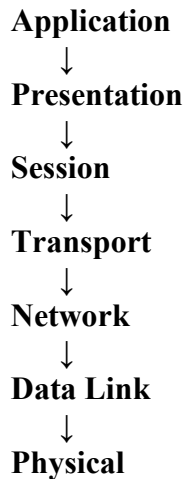
RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well-organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping
Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well-structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

ANSWERS:

1. Missing OSI Layers

Concept Map



Explanation

The missing layers are Session Layer and Data Link Layer.

The Session Layer establishes, manages, synchronizes, and terminates communication sessions between two devices. It inserts synchronization points (checkpoints) so that communication can continue from the last checkpoint if a failure occurs.

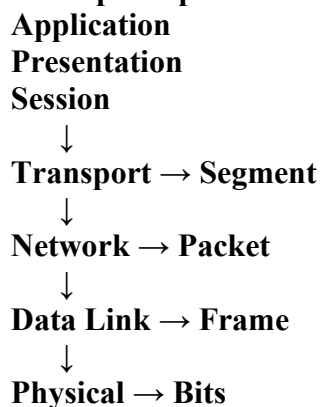
The Data Link Layer provides node-to-node communication by converting packets into frames. It performs framing, physical addressing (MAC address), error detection using CRC, and flow control to ensure reliable transmission over the physical medium.

Analysis

The Session Layer ensures continuous communication by maintaining active sessions, while the Data Link Layer guarantees error-free frame transmission between neighboring devices. Together with the other OSI layers, they provide reliable and efficient end-to-end communication.

2. Data Units and OSI Layers

Concept Map



Explanation

When user data is transmitted, each OSI layer converts it into a different Protocol Data Unit (PDU).

Transport Layer → Segment

Network Layer → Packet

Data Link Layer → Frame

Physical Layer → Bits

During transmission, the sender performs encapsulation, where each layer adds its own header. At the receiver, decapsulation removes these headers in reverse order to recover the original data.

Analysis

The transformation of data units ensures that addressing, routing, error detection, and physical transmission are handled correctly. This layered approach simplifies communication and improves reliability across different network technologies.

3. MAC Address and IP Address

Concept Map

Data Link Layer



MAC Address



Local Delivery

Network Layer



IP Address



Global Delivery

Explanation

The MAC Address is a unique hardware address used for communication within the same Local Area Network (LAN). Switches use MAC addresses to deliver frames between devices.

The IP Address is a logical address used by routers to forward packets between different networks across the Internet.

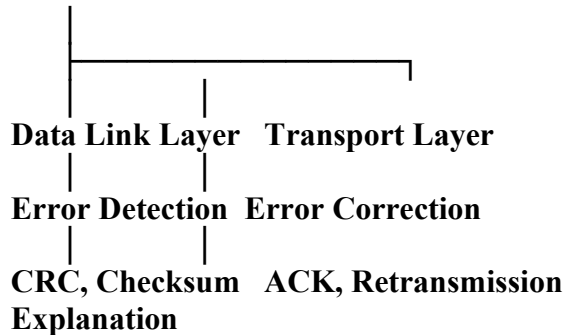
Analysis

When a user sends data to another computer on a different network, the IP address identifies the destination network, while the MAC address identifies the next local device. The cooperation of these two addressing methods enables complete end-to-end communication.

4. Error Control

Concept Map

Error Control



The Data Link Layer detects transmission errors using techniques such as Cyclic Redundancy Check (CRC) and checksum before forwarding frames.

The Transport Layer ensures reliable communication using acknowledgements (ACKs), sequence numbers, flow control, and retransmission of lost segments.

Analysis

Both layers work together to improve reliability. The Data Link Layer detects errors on individual links, while the Transport Layer guarantees complete and correct delivery between the source and destination. This minimizes data loss and improves network performance.

5. Web Browsing Using OSI Layers

Concept Map

Application → HTTP/HTTPS



Presentation → SSL/TLS Encryption



Session → Session Management



Transport → TCP



Network → IP Routing



Data Link → Ethernet/Wi-Fi



Physical → Cable/Fiber/Wireless Signals

Explanation

When a user opens a website, the Application Layer uses HTTP/HTTPS to request the webpage. The Presentation Layer encrypts and decrypts data using SSL/TLS. The Session Layer establishes and maintains communication sessions. The Transport Layer uses TCP to ensure reliable delivery through acknowledgements and retransmissions. The Network Layer routes packets using IP addresses. The Data Link Layer delivers frames within the local network, while the Physical Layer transmits bits through cables or wireless signals.

Analysis

Each layer depends on the services of the layer below it. If any layer fails, communication is

interrupted. For example:

Physical Layer failure → No signal transmission.

Data Link Layer failure → Frames cannot be delivered.

Network Layer failure → Packets cannot reach the destination.

Transport Layer failure → Reliable communication is lost.

Application Layer failure → The webpage cannot be displayed.

The layered architecture simplifies troubleshooting and ensures efficient web communication.

6. Network Troubleshooting

Concept Map

Physical ✓



Data Link ✓



Network ✗



Transport ✗



Application ✗

Explanation

The concept map shows that the Physical Layer and Data Link Layer are functioning correctly because cables, signals, and local frame delivery are working. However, the Network Layer has failed, preventing packets from being routed to the destination. As a result, the Transport Layer cannot establish reliable communication, and the Application Layer cannot provide services to the user.

Possible causes of the Network Layer failure include:

Incorrect IP address configuration

Router failure

Routing table errors

IP address conflicts

Analysis

The concept map helps administrators quickly identify the faulty layer. Since the lower layers are working properly, troubleshooting should begin at the Network Layer by checking IP configuration, routing tables, gateway settings, and router connectivity. This systematic approach reduces troubleshooting time and improves network reliability.